# Base station physics: summary

Params: base/spec/params.py v0.1-stub (still defaults for LAUNCH, AIM, REEL, LAYOUT)

### Spring (each, A228, SF 1.2 on Shigley static limits)

| | |
|---|---|
| Rate used / by formula | 0.125 / 0.131 N/mm |
| Initial tension used | 7.0 N |
| Body coils, mass | 96, 54 g |
| Safe force (governed by) | 43.6 N (hook torsion) |
| Safe extension (wire) / spec cap | 293 / 250 mm |
| Surge (fixed-fixed) | 25 Hz |

### Launch (chain spring model)

| | |
|---|---|
| Stroke, ratio, pitch | 245 mm, 1.0:1, 35 deg |
| Spring extension full / rest | 280 / 35 mm |
| Stored energy (2 springs) | 13.1 J |
| Carriage force at full draw | 84 N |
| Muzzle speed | 7.61 m/s |
| Peak acceleration, stroke time | 21 g, 55 ms |
| Efficiency (dart KE / stored) | 62 % |
| Carriage KE at stop, pad peak | 2.32 J, 434 N (15 mm pad) |
| Spring KE at stop, coil clash | 0.94 J, 8.3 m/s |

### Flight (bead-chain tether + reel model, level ground)

| | |
|---|---|
| Range at 0.5 / 0.75 / 1.0 N brake | 4.93 / 4.46 / 4.04 m (pitch 36) |
| Same shot, no tether | 5.93 m |
| Flight time, apex (0.75 N) | 0.89 s, 1.24 m |
| Impact speed, angle (0.75 N) | 6.2 m/s, 54 deg down |
| Ballistics table | 6 to 47 deg, best 36.0 deg: 4.45 m |

### Storage reel payout (0.75 N at full reel)

| | |
|---|---|
| Reel empty mass, inertia full | 35 g, 19.4 kg mm2 |
| Line radius full / core | 27.4 / 20.0 mm (brake +37 % at core) |
| Line pull in the stroke (spin-up) | 4.37 N |
| Reel rpm max | 2652 rpm |
| Overrun after landing | 0.25 m in 0.13 s |

### Capstan and tow

| | |
|---|---|
| Wraps for tow limit (mu 0.15) | 3.3 raw, 5 design |
| Take-up needed (design wraps) | 0.45 N |
| Tow at limit: torque, speed, current | 2.02 N m, 0.119 m/s, 3.72 A |
| Line pull at limiter (5 N m) (12.6 V) | 127 N (break 667 N) |
| Line pull, motor stall (no limiter, 12.6 V) | 358 N |

### Drivetrain (wiper at 11.1 V with sag)

| | |
|---|---|
| aim: peak torque, current, time | 0.33 N m, 1.9 A, 7 s |
| cock: peak torque, current, time | 3.07 N m, 4.9 A, 3 s |
| reel: peak torque, current, time | 0.17 N m, 1.7 A, 142 s |
| tow: peak torque, current, time | 2.02 N m, 3.7 A, 42 s |
| Wiper at limiter (5 N m), 12.6 V (with sag) | 5.0 N m, 7.0 A |
| Wiper motor stall (no limiter), 12.6 V (with sag) | 14.1 N m, 17.0 A |

### Harry must measure (then edit the parameter block and rerun: python summary.py)

- Spring hang test: hang 1, 2, 3 kg from one spring, measure the length each time. Slope = rate, intercept = initial tension. Every range number scales with the stored energy.
- Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. Edit common.Wiper or spec WIPER.
- Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg).
- Slip clutch: pull the line off the full reel with a luggage scale; set 0.4 to 0.5 N (FINDINGS F5).
- Rail friction: tilt the rail with the dart on it until it slides (mu = tan angle).